LLARS: Enabling Domain Experts & Developers Collaboration for LLM Prompting, Generation and Evaluation

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Abstract

We demonstrate LLARS, an open-source platform that bridges the gap between domain experts and developers for building LLM-based assistants. LLARS integrates three tightly connected modules into an end-to-end pipeline: **Collaborative Prompt Engineering** for real-time co-authoring with version control and instant LLM testing, **Batch Generation** for configurable output production across user-selected prompts × models × data with cost control and **Evaluation** for human and LLM assessors to rate, rank, label or compare outputs with live inter-rater agreement metrics. New prompts and models are automatically available for batch generation and completed batches can be turned into evaluation scenarios with a single click. Interviews with six domain experts and three developers in online counselling confirmed that LLARS eliminates the "translation work" between disciplines and enables domain experts to directly co-author prompts and conduct evaluations independently.

Demo: https://llars-project.github.io/demo

1 Introduction

Organizations building LLM-based assistants for sensitive domains [Chen *et al.*, 2024; Ling *et al.*, 2025]—online counselling [Stade *et al.*, 2024], legal analysis [Katz *et al.*, 2024], medical documentation [Van Veen *et al.*, 2024]—must write effective prompts, select suitable models and rigorously evaluate outputs, requiring both domain expertise and technical skill [Zamfirescu-Pereira *et al.*, 2023; Schulhoff *et al.*, 2024]. Public benchmarks provide useful orientation but rarely transfer to domain-specific tasks; practitioners must test models on their own data to identify the smallest—and most cost-effective—option that still meets quality requirements, especially when replacing a model in a running system. Yet domain experts and developers typically work in isolation: prompts are iterated in shared documents without version control and no single tool supports the full cycle from prompt development through systematic evaluation.

We present LLARS (LLM Assisted Research System), an open-source platform that closes this gap with three integrated modules (Figure 1). *Collaborative Prompt Engineering* lets domain experts and developers co-author prompts in a shared real-time editor with version control and instant LLM testing. *Batch Generation* produces the user-configured Cartesian product of prompts × models × data items with cost estimation and budget control. *Evaluation* lets human and LLM evaluators rate, rank, label or compare outputs with live agreement metrics. The modules are tightly connected: new prompts and newly registered models automatically appear in batch generation and completed batches become evaluation scenarios with a single click. Each module supports JSON/CSV export and connects to any OpenAI-compatible provider.

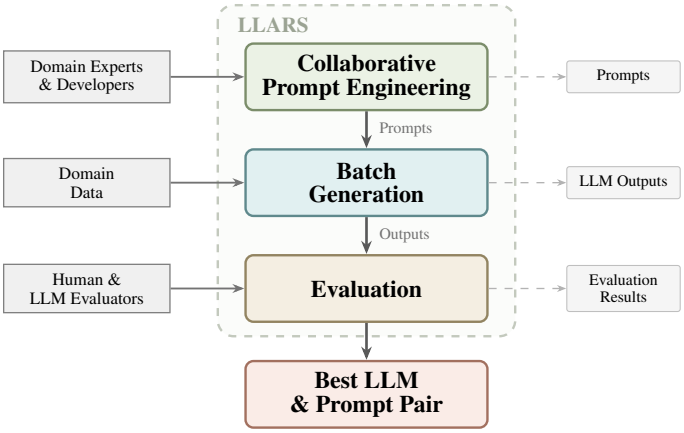

Figure 1: LLARS pipeline: domain experts and developers collaboratively develop prompts, generate outputs across LLMs and evaluate results with human and LLM evaluators. Each stage supports export; the pipeline yields a validated model–prompt combination.

Table 1 surveys twelve existing platforms along these dimensions. Agenta [Agenta AI, 2024] combines prompt versioning with LLM-as-judge scoring and provides A/B deployment testing, but lacks real-time co-editing and structured human evaluation with inter-rater agreement metrics. Phoenix [Arize AI, 2024], Langfuse [Langfuse GmbH, 2024] and W&B Weave [Weights & Biases, 2024] provide observability dashboards with dataset experiments and per-output feedback but focus on tracing and debugging production sys-

Table 1: Comparison of GUI-first tools for prompt engineering, batch generation and hybrid LLM output evaluation. ● = native support; ◐ = partial; ○ = not supported. **Collab.** = real-time collaborative prompt co-editing; **Batch** = systematic generation over prompts × models × data; **H-Eval** = structured human evaluation campaigns; **LLM-E** = LLM-as-evaluator; **IRR** = inter-rater reliability metrics; **E2E** = integrated end-to-end pipeline.

Tool	OSS	Collab.	Batch	H-Eval	LLM-E	IRR	E2E
LLARS	●	●	●	●	●	●	●
Agenta [Agenta AI, 2024]	●	◐	◐	◐	●	○	◐
ChainForge [Arawjo *et al.*, 2024]	●	○	●	○	◐	○	◐
Phoenix [Arize AI, 2024]	●	○	◐	◐	●	○	○
Langfuse [Langfuse GmbH, 2024]	●	○	◐	◐	●	◐	○
W&B Weave [Weights & Biases, 2024]	●	○	◐	◐	●	○	○
Label Studio [HumanSignal, Inc., 2024]	●	○	○	●	◐	◐	○
Argilla [Argilla, Inc., 2024]	●	○	○	●	◐	◐	○
LangSmith [LangChain, Inc., 2024]	○	○	◐	◐	●	○	◐
Braintrust [Braintrust Data, Inc., 2024]	○	○	◐	◐	●	○	◐
Maxim [Maxim AI, 2024]	○	○	◐	◐	●	○	◐
Vellum [Vellum AI, 2024]	○	○	◐	◐	●	○	◐

tems rather than systematic batch generation or coordinated multi-evaluator campaigns. ChainForge [Arawjo *et al.*, 2024] offers a visual node-based interface for multi-model comparison with template variables and supports programmatic evaluation via custom code nodes, but omits structured human evaluation entirely. Label Studio [HumanSignal, Inc., 2024] and Argilla [Argilla, Inc., 2024] offer dedicated human annotation workflows—Label Studio computes Krippendorff's α across annotators, Argilla provides agreement analytics via its API—but both require externally generated outputs, disconnecting evaluation from prompt development. LangSmith [LangChain, Inc., 2024], Braintrust [Braintrust Data, Inc., 2024], Maxim [Maxim AI, 2024] and Vellum [Vellum AI, 2024] provide playground-style testing with dataset-level scoring but are closed-source, lack collaborative editing and omit batch generation. LLARS is an open-source platform to unify collaborative prompt engineering, Cartesian-product batch generation and hybrid human–LLM evaluation with agreement analytics, enabling developers and domain experts to work together from prompt development through rigorous assessment.

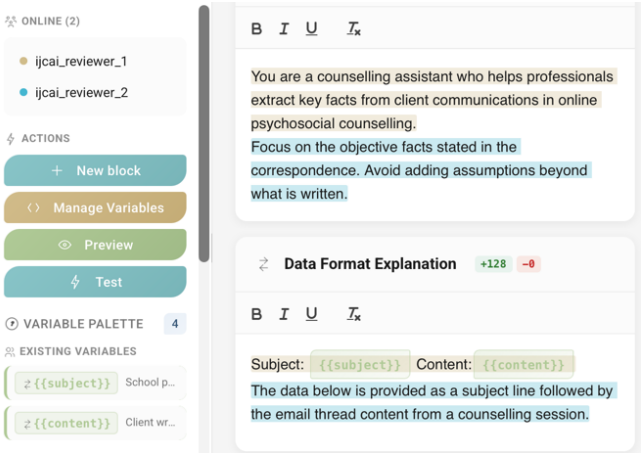

Figure 2: Collaborative prompt editor with a data-format explanation block containing template variables {{subject}} and {{content}}. The sidebar (left) shows actions for block management, variable management, preview and LLM testing; the Variable Palette lists all extracted variables with sample values.

2 Platform Overview

Co-designed with developers, social science researchers and practising counsellors, LLARS runs as a containerized web application for textual data. Teams register proprietary or self-hosted OpenAI-compatible LLM providers; registered models then appear across all three modules.

2.1 Collaborative Prompt Engineering

The prompt engineering module (Figure 2) provides a shared workspace where domain experts and developers co-author prompts in real time using a collaborative editor built on Yjs [Jahns, 2023]; every keystroke is instantly visible to all connected users and online presence indicators show who is currently editing.

A prompt is composed of ordered blocks—each representing one part of the prompt, such as a data format description or an output instruction—that can be freely added, reordered or removed. One block may optionally be marked as a system block, which becomes the LLM's system prompt; all remaining blocks are concatenated into the user prompt. Each block maintains an independent version history that tracks insertions and deletions per save (visible as `+130 / -0` in Figure 2), enabling diff comparison and rollback to any prior revision without affecting other blocks. Blocks may contain template variables written as {{variable_name}}, which act as placeholders for external data that would otherwise clutter the prompt—an entire email thread, for instance, can be captured in a single {{content}} variable, keeping the surrounding instructions concise and letting authors focus on prompt design rather than data formatting. The platform extracts all variables into a shared Variable Palette (Figure 2, bottom-left), where each can be annotated with

a sample value (e.g. {{`subject`}} = *"School problems at home"*). These samples let collaborators preview the assembled prompt and serve as test inputs: any team member can send the current prompt—with samples substituted—to a selected LLM and receive a streamed response, enabling parallel testing across different models. The editor also functions standalone: prompts can be exported and imported as JSON for use in external systems. Once finalized, a prompt passes directly into batch generation, where the same template variables are systematically filled from dataset columns, scaling a single prompt to hundreds of outputs.

2.2 Batch Generation

While the prompt editor allows rapid testing with sample values, a single example cannot validate whether a prompt generalizes across diverse inputs. Batch generation addresses this: users select from available prompts—self-authored or shared by collaborators—and any registered model to apply to their data, producing the Cartesian product of prompts $\times$ models $\times$ data items. A generation matrix previews this combination and estimates cost based on model pricing and projected token counts; optional budget caps pause jobs that exceed a threshold. A wizard offers three data source modes, each suited to a different stage of development: upload data (provide new CSV or JSON input), previous generation (reuse data from a completed batch job) and prompt only (generate without source data).

In upload-data mode, users provide a CSV or JSON file whose columns map to the prompt's template variables—for example, 50 counselling email threads as JSON, each entry containing `subject` and `content` fields matching the variables defined during prompt engineering. Combined with two prompts and two models, LLARS produces $50 \times 2 \times 2 = 200$ outputs. Previous-generation mode works identically but reuses the data from a completed batch job, so that new prompts or additional models can be tested on exactly the same dataset without re-uploading. Prompt-only mode sends selected prompts to selected models without uploading data; if the prompt contains variables, the sample values from prompt engineering are used.

Results stream in real time and every output records its full provenance—source item, prompt version, model, parameters, token count and cost—exportable as CSV or JSON. This provenance is central to the pipeline: when outputs reach evaluation, assessors see only the generated text without any indication of which model or prompt produced it and the platform maps their judgments back to specific model–prompt combinations, directly informing selection. Completed batches transfer into evaluation with a single click, preserving all attribution.

2.3 Evaluation

The evaluation module brings human and LLM evaluators together in a single framework. Four general evaluation types are available (Figure 3): *rating* with configurable multi-dimensional Likert scales, *ranking* into ordered buckets, *labeling* for binary or multi-class classification and pairwise *comparison*. The type system is extensible: for the counselling domain, two additional types were added—*mail rating* for assessing email thread quality and *authenticity* for distinguishing human-written from LLM-generated text (Figure 3). An AI-assisted Scenario Wizard analyzes uploaded data to recommend the most suitable type; scenarios can also be created directly from batch generation outputs, preserving model attribution.

Scenario owners control how items are distributed—all items to every evaluator for full inter-rater analysis or configurable subsets for faster throughput. Where most tools limit LLMs to pairwise judging [Zheng *et al.*, 2024], LLARS treats them as full evaluation participants that rate, rank, label and compare alongside humans using customizable type-specific prompts. Agreement metrics update live—Krippendorff's α, Cohen's and Fleiss' κ, Kendall's τ and ICC—filterable by evaluator pairing (human–human, LLM–LLM, human–LLM). All data, metrics and configurations are exportable as JSON or CSV.

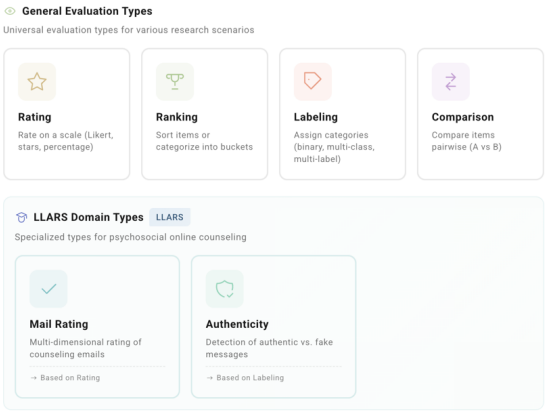

Figure 3: AI-assisted Scenario Wizard: task selection step with four general and two domain-specific evaluation types. The wizard analyzes uploaded data and recommends a suitable type.

3 Deployment and User Study

LLARS is actively used in online counselling research: the Virtual Client project [Rudolph *et al.*, 2024] uses it to develop prompts for simulated client interactions and CAIA [Steigerwald *et al.*, 2025], a co-designed AI assistant for online counselling, relies on it for prompt development and evaluation. A full end-to-end case study [Steigerwald and Albrecht, 2025] produced 253 subject lines across 11 LLMs and had five professionals with an LLM evaluator rate them (1,518 assessments)—determining how small a model could be while still producing meaningful results.

Semi-structured interviews with six domain experts and three developers confirmed that consolidating the pipeline in one platform saves considerable time. Domain experts found prompt authoring straightforward—particularly for self-hosted models lacking external playgrounds—and rated evaluation as immediately understandable; developers valued that prompts, data and outputs remain in one place. Both groups emphasised that co-locating the workflow removed the "translation work" between disciplines.

4 Demo and Conclusion

Our demo walks attendees through the complete workflow with pre-built prompts, generation jobs and evaluation scenarios for hands-on exploration. By unifying prompt engineering, batch generation and evaluation, LLARS bridges the gap between domain experts and developers for interdisciplinary research.

References

[Agenta AI, 2024] Agenta AI. Agenta: Open-source LLMOps platform for prompt management, evaluation, and observability. https://agenta.ai, 2024. Accessed: 2026-02-01.

[Arawjo *et al.*, 2024] Ian Arawjo, Chelse Swoopes, Priyan Vaithilingam, Martin Wattenberg, and Elena Glassman. ChainForge: A visual toolkit for prompt engineering and LLM hypothesis testing. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM, 2024.

[Argilla, Inc., 2024] Argilla, Inc. Argilla: Collaboration tool for AI engineers and domain experts. https://argilla.io, 2024. Accessed: 2026-01-06.

[Arize AI, 2024] Arize AI. Phoenix: Open-source AI observability and evaluation. https://phoenix.arize.com, 2024. Accessed: 2026-02-01.

[Braintrust Data, Inc., 2024] Braintrust Data, Inc. Braintrust: AI evaluation and observability platform. https://www.braintrust.dev, 2024. Accessed: 2026-02-01.

[Chen *et al.*, 2024] Zhiyu Zoey Chen, Jing Ma, Xinlu Zhang, Nan Hao, An Yan, Armineh Nourbakhsh, Xianjun Yang, Julian McAuley, Linda Petzold, and William Yang Wang. A survey on large language models for critical societal domains: Finance, healthcare, and law. *Transactions on Machine Learning Research*, 2024.

[HumanSignal, Inc., 2024] HumanSignal, Inc. Label studio: Open source data labeling platform. https://labelstud.io, 2024. Accessed: 2026-01-06.

[Jahns, 2023] Kevin Jahns. Yjs: A CRDT framework for shared editing. https://yjs.dev, 2023. Accessed: 2026-01-06.

[Katz *et al.*, 2024] Daniel Martin Katz, Michael James Bommarito, Shang Gao, and Pablo Arredondo. GPT-4 passes the bar exam. *Philosophical Transactions of the Royal Society A*, 382(2270):20230254, 2024.

[LangChain, Inc., 2024] LangChain, Inc. LangSmith: AI agent evaluation platform. https://www.langchain.com/langsmith, 2024. Accessed: 2026-02-01.

[Langfuse GmbH, 2024] Langfuse GmbH. Langfuse: Open-source LLM engineering platform. https://langfuse.com, 2024. Accessed: 2026-02-01.

[Ling *et al.*, 2025] Chen Ling, Xujiang Zhao, Jiaying Lu, Chengyuan Deng, Can Zheng, Junxiang Wang, Tanmoy Chowdhury, Yun Li, Hejie Cui, Xuchao Zhang, et al. Domain specialization as the key to make large language models disruptive: A comprehensive survey. *ACM Computing Surveys*, 58(3), 2025.

[Maxim AI, 2024] Maxim AI. Maxim: GenAI evaluation and observability platform. https://www.getmaxim.ai, 2024. Accessed: 2026-02-01.

[Rudolph *et al.*, 2024] Eric Rudolph, Natalie Engert, and Jens Albrecht. An AI-based virtual client for educational role-playing in the training of online counselors. In *Proceedings of the 16th International Conference on Computer Supported Education (CSEDU)*, pages 108–117, 2024.

[Schulhoff *et al.*, 2024] Sander Schulhoff, Michael Ilie, Nishant Balepur, Konstantine Kahadze, Amanda Liu, Chenglei Si, Yinheng Li, Aayush Gupta, HyoJung Han, Sevien Schulhoff, et al. The prompt report: A systematic survey of prompting techniques. *arXiv preprint arXiv:2406.06608*, 2024.

[Stade *et al.*, 2024] Elizabeth C. Stade, Shannon Wiltsey Stirman, Lyle H. Ungar, Cody L. Boland, H. Andrew Schwartz, David B. Yaden, Joao Sedoc, Robert J. DeRubeis, Robb Willer, and Johannes C. Eichstaedt. Large language models could change the future of behavioral healthcare: A proposal for responsible development and evaluation. *npj Mental Health Research*, 3:12, 2024.

[Steigerwald and Albrecht, 2025] Philipp Steigerwald and Jens Albrecht. Comparing large language models for automated subject line generation in e-mental health: A performance study. In *Proceedings of the 11th International Conference on Information and Communication Technologies for Ageing Well and e-Health (ICT4AWE)*, pages 70–77. SciTePress, 2025.

[Steigerwald *et al.*, 2025] Philipp Steigerwald, Nico Bienlein, Jennifer Burghardt, Mara Stieler, Robert Lehmann, and Jens Albrecht. CAIA in practice: Field evaluation of an AI-assisted support system for text-based online counselling. In *Proceedings of the IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2025. Accepted.

[Van Veen *et al.*, 2024] Dave Van Veen, Cara Van Uden, Louis Blankemeier, Jean-Benoit Delbrouck, Asad Aali, Christian Bluethgen, Anuj Pareek, Malgorzata Polacin, Eduardo Pontes Reis, Anna Seehofnerova, et al. Adapted large language models can outperform medical experts in clinical text summarization. *Nature Medicine*, 30:1134–1142, 2024.

[Vellum AI, 2024] Vellum AI. Vellum: AI application development platform. https://www.vellum.ai, 2024. Accessed: 2026-02-01.

[Weights & Biases, 2024] Weights & Biases. Weave: Toolkit for developing AI-powered applications. https://wandb.ai/site/weave, 2024. Accessed: 2026-02-01.

[Zamfirescu-Pereira *et al.*, 2023] J.D. Zamfirescu-Pereira, Richmond Y. Wong, Bjoern Hartmann, and Qian Yang. Why johnny can't prompt: How non-AI experts try (and fail) to design LLM prompts. In *Proceedings of the*

2023 CHI Conference on Human Factors in Computing Systems. ACM, 2023.

[Zheng *et al.*, 2024] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2024.